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Apollo 11 Mission Overview

Relatório Intermédio

Palavras Chave

texto livro, arquitetura, história, construção, materiais de construção, saber tradicional.

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Chapter 1

Introduction

1.1 Background

Context or background of the project.

1.2 Problem Statement

Clearly state the problem that you will address in your project.

1.3 Objectives

Clearly state the objectives or goals of your project.

1.4 Significance of the Study

Explain the importance of your project.

1.5 Scope

Define the scope of your project through requirements and allocate priorities in accordance with the relevance and uncertainty of each one.

1.6 Work Breakdown Structure (WBS)

Define the WBS, including Work Packages (WPs), tasks, Gantt Chart, indentifying risks and mitigation strategy.

Chapter 2

Literature Review

2.1 Thermodynamic Basis for Corrosion

Corrosion is fundamentally a spontaneous thermodynamic process driven by the tendency of metals to revert to their most stable, lower-energy states, such as oxides (M_nO_p) or sulfides (M_nS_p) [20, 18, 21]. This transition is quantified by a negative change in Gibbs free energy (ΔG), indicating that the oxidation reaction is energetically favoured over the retention of the metallic state. Mathematically, this thermodynamic driving force is related to the electrochemical potential E through the relation:

$$\Delta G = -nFE \tag{2.1}$$

where n is the number of electrons transferred per mole of reaction and F is the Faraday constant[20, 7, 18]. A negative ΔG corresponds to a positive E, confirming that the reaction proceeds spontaneously in the direction of corrosion.

2.2 Notion of Potential and Passivity

In practice, the thermodynamic behaviour of a metal is governed by its electrochemical potential in a given environment. While standard electrode potentials (E°) provide a useful thermodynamic reference, they are defined under idealised conditions (unit activity, 298 K, 1 atm) and therefore have limited direct applicability to real corrosion environments. Corrosion studies consequently rely on the open-circuit potential (OCP), defined as the potential difference between the metal and a reference electrode measured in the absence of any externally applied current [21, 19]. Since absolute electrode potentials cannot be determined experimentally, OCP values are always reported relative to a specified reference electrode in a given electrolyte [21, 20, 18].

The OCP must be carefully distinguished from E°, the latter representing the ideal thermodynamic driving force for a given half-reaction under standard conditions. For aluminium, $E^\circ = -1.66$ V versus the standard hydrogen electrode (SHE), which serves as the universal 0 V reference. In contrast, the OCP reflects the steady-state potential established in a specific environment, at which the rates of anodic and cathodic partial reactions are equal and no net current flows [20, 18, 14]. In aqueous environments, aluminium typically exhibits an OCP in the range of -0.5 V to -0.7 V versus SHE [21, 18]. This substantial positive shift relative to E° arises from the rapid and spontaneous formation of a thin, adherent alumina film upon exposure to air or aqueous media:

$$2\,\mathrm{Al} + \frac{3}{2}\,\mathrm{O_2} \rightarrow \mathrm{Al_2O_3} \tag{2.2}$$

This native passive oxide layer acts as a physical barrier between the metal and the electrolyte, significantly retarding further oxidation and shifting the OCP in the noble direction. The stability of this passive film, however, is strongly dependent on environmental conditions, particularly pH, temperature, and the presence of aggressive anions such as chloride [21, 20, 7, 18].

2.3 Pourbaix Diagrams

A powerful tool for visualising the thermodynamic stability of a metal as a function of both electrochemical potential and pH is the Pourbaix diagram, also known as the E–pH diagram. For a given metal–electrolyte system, the diagram delineates three distinct thermodynamic domains [18, 7]:

- **Corrosion**, where the stable species are soluble ionic products (e.g. Al^{3+}, AlO_2^-), indicating active dissolution of the metal;
- **Passivation**, where the stable species are insoluble oxides or hydroxides (e.g. Al_2O_3, $Al(OH)_3$) that form a protective layer on the metal surface;
- **Immunity**, where the metal itself is the thermodynamically stable phase and dissolution is thermodynamically unfavourable.

The E–pH diagram for aluminium, shown in [REFERENCIAR FIGURE], illustrates the amphoteric nature of the metal: aluminium is susceptible to corrosion in both strongly acidic and strongly alkaline media, with only a relatively narrow intermediate pH range — approximately pH 4 to pH 8.5 — corresponding to the passivation domain

[21, 7, 18]. This behaviour is consistent with the dissolution reactions at either extreme:

$$\mathrm{Al} + 3\,\mathrm{H}^+ \rightarrow \mathrm{Al}^{3+} + \frac{3}{2}\,\mathrm{H}_2 \quad \text{(acidic dissolution)} \tag{2.3}$$

$$\mathrm{Al} + \mathrm{OH}^- + \mathrm{H}_2\mathrm{O} \rightarrow \mathrm{AlO}_2^- + \frac{3}{2}\,\mathrm{H}_2 \quad \text{(alkaline dissolution)} \tag{2.4}$$

It must be emphasised, however, that Pourbaix diagrams are strictly equilibrium constructs: they identify thermodynamically stable phases and the direction of spontaneous reactions, but provide no information on the kinetics of corrosion or the rate at which these reactions proceed [18, 21, 20]. A metal situated within the passivation domain may still corrode if the passive film is mechanically damaged or chemically disrupted [7, 18]. An example of this is when aluminium is exposed to chloride-containing environments, where localised breakdown of the passive film can initiate pitting corrosion despite nominally passive conditions [21, 20].

2.4 Kinetics of Corrosion

The kinetics of corrosion are fundamentally governed by the rate of charge transfer at the interface between a metal and its surrounding aqueous environment [21]. During corrosion, metal atoms are oxidised at anodic sites, releasing electrons into the metal lattice, while electroactive species in solution simultaneously capture these electrons at cathodic sites [21]. These two partial reactions — the anodic oxidation and the cathodic reduction — are spatially and electrically coupled, and their relative rates determine the overall corrosion rate.

This charge exchange is confined to a highly structured interfacial region, approximately 10 nm in thickness, known as the electrical double layer [21]. Upon immersion of a metal in an electrolyte, the perturbation of the local molecular and ionic arrangement gives rise to a charge separation at the interface, which is sustained to maintain overall electroneutrality [21, 20]. The structure of the double layer is described by a combination of the Helmholtz, Stern, and Gouy–Chapman models, which together delineate three distinct regions [21, 20]:

- **Compact Stern Layer (Inner Helmholtz Plane):** This innermost region, extending approximately 3 Å from the surface, comprises oriented water molecules and specifically adsorbed species — most notably desolvated anions such as chloride, which have shed their hydration shell to bond directly to the metal surface [21, 20];

- **Outer Helmholtz Plane:** Situated approximately 5 to 6 Å from the surface, this plane marks the closest approach of solvated ionic species that retain their hydration shell [21, 20];

- **Diffuse Gouy–Chapman Layer:** Extending from the outer Helmholtz plane into the bulk electrolyte, this region consists of a diffuse ionic distribution whose thickness, characterised by the Debye length, decreases with increasing ionic strength [21, 20].

The architecture of the double layer directly controls corrosion kinetics by imposing physical and electrostatic barriers to ionic transport and charge transfer [20]. These barriers give rise to polarisation — a measurable retardation of the electrochemical reactions — which ultimately governs the practical rate at which a metal corrodes [21, 20].

In the context of equivalent circuit modelling, the double layer is represented as a capacitance in parallel with a charge-transfer resistance, where the capacitive element reflects the charge separation across the interface and the resistive element reflects the activation barrier to Faradaic reactions [20].

2.5 Influence of IMPs on LDH growth

AC46000 contains a diverse range of intermetallic particles (IMPs), including copper-rich phases such as Al_2Cu and Al_7Cu_2Fe, iron-rich phases such as Al_5FeSi and $Al_{15}(Fe, Mn)_3Si_2$, and eutectic silicon particles. To assess the influence of these IMPs on LDH growth and corrosion behavior, it is important to consider their electrochemical potential relative to the aluminium matrix [8, 10].

The copper-rich phases are strongly cathodic relative to the Al matrix, with Al_2Cu exhibiting a potential difference of approximately +400 mV [8]. This drives intense localized micro-galvanic coupling, promoting rapid oxygen reduction and a sharp rise in local pH. In the context of LDH growth, these conditions trigger the preferential and rapid precipitation of thick, "flower-like" LDH islands directly over the cathodic sites [3]. This localized growth is sustained by the anodic dissolution of the surrounding aluminium matrix, commonly referred to as trenching [2]. The trenching process generates voids at the periphery of the intermetallic particles where LDH flakes fail to nucleate [2]. These uncoated gaps compromise the corrosion resistance of the substrate, as chloride ions and other aggressive species can readily penetrate the exposed defects and initiate pitting corrosion [2]. The resulting inhomogeneous LDH film also reduces the adhesion of subsequently applied polymer coatings or paints [2], and in severe cases extensive trenching can render the intermetallic particles

prone to detachment, further undermining the integrity of overlying layers [2].

The iron-rich phases are also strongly cathodic, approximately +300 mV above the aluminium matrix [8], and the galvanic coupling they generate similarly leads to inhomogeneous LDH growth and increased susceptibility to corrosion.

Although eutectic silicon particles are likewise cathodic, with a potential difference of approximately +250 mV relative to the Al matrix, they are electrochemically inert and do not act as nucleation sites for LDH. Since they provide no source of the metal cations (Al^{2+}/Al^{3+}, Mg^{2+}) required for LDH formation, no coating growth is triggered at these sites [8, 6].

2.5.1 Pre-Treatment

Given the detrimental influence of these IMPs on LDH growth homogeneity and overall coating quality, their removal from the alloy surface prior to conversion coating is essential. To this end, various pre-treatment strategies have been developed, but the most widely adopted approach consists of a multistep procedure comprising degreasing, alkaline etching, and acid pickling, applied sequentially [15].

Degreasing constitutes the first pre-treatment step, with the primary objective of removing surface contamination, most notably oils and lubricants introduced during the manufacturing process. This step is critical to ensure full surface wettability, thereby guaranteeing that subsequent processing stages act uniformly across the substrate [15].

The second step, alkaline etching, is aimed at dissolving the native oxide layer that forms spontaneously on aluminium surfaces, as well as removing surface-exposed intermetallic particles. The cathodic IMPs act as highly active sites for oxygen reduction, generating a localised rise in pH at their immediate vicinity. This accelerates the anodic dissolution of the surrounding aluminium matrix, progressively undermining the mechanical support of the particles [17]. As the matrix recedes, scallop-shaped voids form at the particle periphery, eventually causing the IMPs to detach from the surface, a process known as selective dealloying [17, 9]. Since many of the constituent alloying elements released during this process are insoluble in alkaline media, they accumulate on the surface as a loosely adherent residue known as smut, the removal of which necessitates a subsequent acid pickling step [15, 17, 9].

Nitric acid (HNO_3) is widely used for desmutting because it naturally passivates the aluminum matrix while acting as a powerful oxidant [9]. It successfully targets and dissolves the dealloyed metallic smut, including copper, iron, and manganese residues. Despite this, Nitric acid alone is ineffective against silicon-rich intermetallic residues [4]. To remove these, fluorides (such as hydrofluoric acid or ammonium bifluoride) must be

added to the pickling bath [1, 13]. The fluoride ions dissolve the silica and complex the remaining silicon remnants so they can be washed away [1, 13].

2.6 Equivalent Circuit Models for EIS

2.6.1 Randles Equivalent Circuit (REC)

Determining an appropriate equivalent circuit model (ECM) for electrochemical impedance spectroscopy (EIS) spectra is among the most critical steps in impedance analysis, as it enables the deconvolution of overlapping electrochemical processes and their quantification through physically meaningful parameters. One of the most widely used models is the Randles equivalent circuit, which consists of a solution resistance (R_{s}) in series with a parallel combination of a charge-transfer resistance (R_{ct}) and a double-layer capacitance (C_{dl}), as represented in [REFERENCIAR FIGURE] [5, 12].

Before discussing each circuit element individually, it is necessary to introduce the concept of the time constant, as it underpins the physical interpretation of the impedance response. A time constant (τ) defines the characteristic duration required for a given electrochemical process to relax from a transient state towards a new equilibrium [5]. In an electrical equivalent circuit, it is defined as the product of a resistance and a capacitance ($\tau = RC$), and governs the rate at which the system responds to an applied perturbation [12]. Because an electrochemical interface involves multiple concurrent processes (such as charge transfer, ionic diffusion, and intermediate adsorption) each process is characterised by its own distinct time constant [5]. For instance, the kinetic time constant of an interface is given by $\tau_k = R_{\mathrm{ct}}C_{\mathrm{dl}}$ [12]. In the frequency domain, each time constant manifests as a characteristic angular frequency ($\omega_c = \frac{1}{RC}$), which marks the transition between different impedance regimes — for example, from a capacitively controlled response to a diffusion-dominated one [12, 5]. Their resolution in the impedance spectrum is therefore a prerequisite for the unambiguous assignment of ECM elements to specific interfacial processes.

R_{s}, also referred to as the ohmic or uncompensated resistance, quantifies the resistive potential drop occurring between the working electrode and the reference electrode [16, 12]. Its magnitude is governed by the specific conductivity of the bulk electrolyte and the cell geometry [16]. In a Nyquist representation, R_{s} is readily identified as the high-frequency intercept of the impedance data with the real axis, corresponding to the condition under which the reactive elements of the circuit are effectively short-circuited and the purely resistive contribution dominates the response.

R_{ct} is the real component of the Faradaic impedance governing the

transfer of electrons across the electrode/electrolyte interface to or from electroactive species in solution [12]. As a strictly kinetic parameter, its value is inversely proportional to the heterogeneous rate constants of the electrode reactions, and it is sometimes referred to as polarisation resistance in the context of Faradaic processes [12, 5]. This inverse relationship extends directly to the corrosion rate: according to the Stern–Geary relationship, the instantaneous corrosion current density i_{corr} is obtained by dividing a proportionality constant B by R_{ct} [11]:

$$i_{\text{corr}} = \frac{B}{R_{\text{ct}}} \tag{2.5}$$

Consequently, a higher R_{ct} reflects slower interfacial kinetics, a lower corrosion current density, and therefore improved corrosion protection [11]. In the Nyquist plot, R_{ct} manifests as the diameter of the semicircle observed in the high-frequency to mid-frequency range, a feature arising directly from its coupling with C_{dl} to form the kinetic time constant $\tau_k = R_{\text{ct}}C_{\text{dl}}$ [5, 12].

C_{dl} is the capacitance that inherently forms at the electrode/electrolyte interface as a consequence of the charge separation between ionic species in solution and charges within the electrode, as described in section [REFERENCIAR SECÇÃO DA DOUBLE LAYER] [11, 16]. While liquid electrodes can behave as nearly ideal capacitors, solid electrodes invariably exhibit frequency dispersion due to surface heterogeneities, roughness, porosity, and non-uniform current distributions [12, 5]. To account for this non-ideal behaviour, the double layer at solid electrodes is most accurately represented by a constant phase element (CPE) rather than an ideal capacitor [5, 12, 16].

From a corrosion monitoring perspective, C_{dl} is a particularly sensitive indicator of coating degradation and the onset of localised pitting, often preceding any macroscopic visual evidence of corrosion [11]. As a protective coating degrades or delaminates, the electrolyte gains direct access to the underlying metal substrate, initiating underfilm corrosion [11]. The measured capacitance at defect sites scales directly with the delaminated area. Consequently, progressive coating breakdown exposes an increasingly large area of bare metal, driving C_{dl} proportionally higher [11]. This area dependence is readily understood through the expression for a planar capacitor:

$$C = \frac{\epsilon\epsilon_0 A}{d} \tag{2.6}$$

where ϵ is the relative permittivity of the interfacial medium, ϵ_0 is the permittivity of free space, A is the effective interfacial area, and d is the charge separation distance. Additionally, as localised corrosion progresses, for instance through pitting, the electrode surface becomes increasingly

rough and geometrically complex, further enlarging the effective interfacial area and driving C_{dl} to higher values [12, 11].

2.6.2 More advanced circuits

While the simple Randles circuit provides an adequate description of a bare metal undergoing a single-step charge transfer reaction, real-world systems exhibit multiple time constants that cannot be captured by a single RC network. To model these more complex responses, advanced equivalent circuits are constructed by combining multiple resistor–capacitor (RC) or resistor–constant phase element (R-CPE) sub-networks.

When a system displays two distinct time constants, which typically manifest as two partially or fully resolved semicircles in the Nyquist plot, the two most common topologies are the series (Voigt-type) configuration [FIGURE X] and the nested (ladder) configuration [FIGURE Y]. In the series configuration, each RC sub-network represents a physically distinct and sequential process that operates independently of the others [12, 11]. The nested configuration, by contrast, is generally preferred in corrosion science for the description of defective or porous protective coatings, where simultaneous and spatially overlapping processes occur [11]. In this topology, the outer RC network represents the macroscopic response of the coating itself which can be characterised by the film capacitance and the film resistance. The inner nested RC network on the other hand describes the actively corroding metal/electrolyte interface at the base of the pores, represented by the double-layer capacitance and the charge-transfer resistance [11].

It is important to note, however, that despite their distinct physical interpretations, a series Voigt circuit and a nested ladder circuit can be mathematically equivalent, producing identical impedance values and indistinguishable complex plane plots [12]. The selection of the nested topology must therefore be justified by the physical reality of the system under investigation rather than by mathematical preference alone [12].

In this nested configuration, the outer time constant is associated with the response of the protective film such as an externally applied organic coating, a native oxide layer, or a conversion coating like an LDH film.

The film resistance, R_{film} plays an analogous role to R_{ct} in the sense that both manifest as semicircle diameters in the Nyquist plot and both inversely reflect the degree of protection afforded at their respective interface. However, while R_{ct} is a kinetic parameter governing the rate of electron transfer across the electrode/electrolyte interface, R_{film} is an ionic transport parameter [11]. It represents the resistance to ionic current flow both through the bulk of the film itself and through any microscopic pores and defects connecting the bulk electrolyte to the metal/film interface [11]. Just as a high R_{ct} indicates sluggish interfacial reaction kinetics, a high

R_{film} restricts ionic transport across the film and is therefore directly associated with superior corrosion protection [12, 11].

The film capacitance, C_{film}, is equally informative and serves as a sensitive early indicator of coating degradation, often responding before any macroscopic evidence of corrosion becomes apparent. Its evolution is most directly governed by water uptake: upon immersion in an electrolyte, water progressively penetrates the pores of the coating [12]. Since water possesses a substantially higher dielectric constant (approximately 80) relative to that of a typical coating, its ingress causes a measurable increase in C_{film} [16, 12]. Monitoring this increase is of practical importance, as water uptake is frequently a precursor to active substrate corrosion [11]. More generally, a progressive increase in C_{film} over time is indicative of ongoing coating degradation and a reduction in protective performance.

Chapter 3

Methodology

3.1 Determine ECM for EIS

Determining an appropriate equivalent circuit model (ECM) for electrochemical impedance spectroscopy (EIS) spectra is among the most important steps in impedance analysis, as it enables the deconvolution of overlapping electrochemical processes and their quantification through physically meaningful parameters, such as charge transfer resistance (R_{ct}), film resistance (R_{film}), and double-layer capacitance (C_{dl}).

Despite its centrality to the analysis, ECM identification remains a laborious task. Commercial software packages such as ZView and Gamry Echem Analyst rely on nonlinear least-squares (NLLS) fitting routines, most commonly the Levenberg-Marquardt algorithm, which are inherently sensitive to the choice of initial parameter estimates. While heuristic approaches exist to guide these estimates, such as reading characteristic resistances from the diameters of semicircular arcs in the Nyquist representation or extracting time constants from Bode phase plots, the process remains largely manual and iterative, demanding considerable operator experience to avoid convergence to local minima or physically unreasonable solutions.

In response to these limitations, a number of open-source, programmatic alternatives have been developed, predominantly in Python. Among these, `impedance.py` provides a flexible framework for ECM fitting and validation. More recently, tools such as `AutoECM` and `AutoEIS` have incorporated machine learning methodologies into the workflow. AutoECM employs supervised classification models such as Random Forest, XGBoost (XGB), and convolutional neural network (CNN) architectures. These are trained on libraries of previously validated circuit-spectrum pairs to predict candidate ECMs for new datasets. AutoEIS, by contrast, adopts a probabilistic framework grounded in Bayesian inference, enabling uncertainty quantification in both circuit structure selection and parameter

estimation. A notable limitation shared by both tools, however, is the apparent discontinuation of active maintenance by their respective developers. As a consequence, dependency incompatibilities with more recent versions of core scientific Python libraries have rendered them largely non-functional in current environments.

Given this landscape, `impedance.py` emerges as the most viable open-source option. In its standard implementation, it operates analogously to commercial software: the user must supply the impedance dataset, an *a priori* circuit string, and initial parameter guesses. Two optional fitting configurations are available. The first, `weight_by_modulus`, applies the modulus of the complex impedance as a frequency-dependent weighting factor which ensures that data points spanning several decades of impedance magnitude contribute proportionally to the objective function. The second, `global_opt`, invokes a basin-hopping algorithm, which combines stochastic perturbation of parameter coordinates with local minimization and a Metropolis-type acceptance criterion to escape local minima. Importantly, `global_opt` is incompatible with `weight_by_modulus` and given that EIS datasets span several decades in frequency, commonly from 10 mHz to 100 kHz, and correspondingly large ranges in impedance magnitude, modulus weighting is generally considered essential to prevent low-frequency, high-impedance data from disproportionately dominating the fit residuals.

To overcome the dependence on user-supplied initial guesses while retaining the precision of local optimization, a hybrid fitting framework was developed building upon impedance.py and the Differential Evolution (DE) algorithm. DE is a stochastic, population-based global optimization metaheuristic well-suited to high-dimensional, non-differentiable, and multimodal objective functions. The algorithm evolves a population of candidate solutions across successive generations through three operators: mutation, which generates trial vectors via linear combinations of population members; crossover, which probabilistically exchanges parameter values between candidate and trial vectors; and selection, which retains whichever vector yields a lower objective function value. Crucially, DE requires no gradient information and treats the cost function as a black box, making it particularly appropriate for the complex, rugged parameter landscapes characteristic of multi-element ECM fitting.

In the proposed workflow, the user supplies only physically motivated bounds for each circuit element rather than precise initial estimates. The DE algorithm then explores this parameter space to identify the order of magnitude and approximate value of each component. To accelerate convergence, given that circuit element values typically span many orders of magnitude, the optimization is performed in logarithmic space, transforming the search domain such that each decade is treated uniformly. This logarithmic reparameterization substantially improves the efficiency of

the population-based search and reduces sensitivity to the scale disparity between, for example, charge-transfer resistances on the order of kilohms and double-layer capacitances on the order of microfarads. Once the DE stage yields a sufficiently accurate global estimate, the resulting parameter vector is passed as the initial condition to the Levenberg-Marquardt local optimizer within `impedance.py`, with `weight_by_modulus` activated, to refine the solution and obtain statistically meaningful parameter estimates together with their associated uncertainties.

3.2 Pre-treatments

As previously discussed, the presence of intermetallic phases (IMPs) on the surface of the aluminum alloy can negatively affect LDH growth and must therefore be removed prior to coating. Two different pre-treatment procedures were tested: a chemical treatment and an industrial treatment.

The first procedure consisted of cleaning at room temperature for 120 s in a 0.1 M NaOH solution, followed by etching for 480 s in a 0.1 M HNO_3 solution. Each step was followed by rinsing with deionized water and final drying in air.

The second procedure consisted of cleaning in Metaclean T2001 (alkaline cleaner based on metal silicates with complexing and surface-active agents) at approximately 65 °C for 25 min, followed by rinsing with deionized water. The specimens were then etched in TurcoLiquid Aluminetch N2 (sodium hydroxide and phosphate-based solution) at 65 °C for 45 s, rinsed with deionized water, and desmutted in TurcoLiquid Smutgo NC (based on ferric sulfate, nitric acid, sodium hydrogendifluoride, and sulfuric acid) at 30 °C for approximately 5 min, before a final rinse with deionized water and drying in air.

Chapter 4

Results

Present the results of your project clearly and concisely, using tables and figures where necessary.

Chapter 5

Discussion

Interpret the results and discuss their implications. Compare your findings with previous research. Discuss any limitations of your study.

Chapter 6

Conclusion & Future Work

Sum up the main findings and their importance, and discuss how they have the WBS going forward.

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